

TURINSK, Russian Federation

WMO: 282550

Lat: 58.050N

Lon: 63.683E

Elev: 337

StdP: 14.52

Time Zone: 5.00 (E05)

Period: 94-17

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.7	-22.2	-32.1	0.9	-26.6	-27.3	1.2	-22.0	20.2	16.2	17.6	17.1	3.2	260	0.590

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.9	85.2	69.7	82.0	68.1	78.6	66.1	71.9	82.1	69.9	79.5	67.7	76.4	4.3	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.2	104.8	78.4	66.2	97.8	75.8	64.3	91.2	72.6	35.9	82.0	34.1	79.6	32.3	76.6	77.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.1	14.4	12.5	DB	-30.8	90.5	5.8	2.3	-35.0	92.2	-38.4	93.6	-41.7	94.9	-45.9	96.6	(4)
(5)			WB	-31.0	74.5	5.8	1.9	-35.2	75.8	-38.6	76.9	-41.8	78.0	-46.0	79.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	35.5	3.3	7.5	23.2	38.8	52.2	61.2	64.7	60.0	49.6	37.2	19.2	7.3		
	DBStd	24.19	13.84	13.71	9.28	10.03	9.51	8.12	7.30	7.43	8.27	9.21	13.88	14.71		
	HDD50	6697	1447	1191	831	354	92	10	1	5	108	408	925	1325		
	HDD65	10962	1912	1611	1296	785	409	165	98	192	467	862	1375	1790		
	CDD50	1403	0	0	0	19	159	346	457	316	95	11	0	0		
	CDD65	193	0	0	0	0	11	51	89	38	4	0	0	0		
	CDH74	1924	0	0	0	4	168	527	859	332	34	0	0	0		
	CDH80	503	0	0	0	0	26	146	234	93	4	0	0	0		
	WSAvg	4.9	5.1	5.0	5.4	5.5	5.4	4.4	3.4	3.9	4.6	5.5	5.4	5.2		
	PrecAvg	21.5	1.2	0.9	1.0	1.2	1.9	2.6	2.8	3.1	2.3	1.7	1.5	1.3		
(15) Precipitation	PrecMax	30.0	2.3	2.2	2.5	2.8	4.7	5.2	6.1	7.5	4.2	3.7	2.8	2.6		
	PrecMin	16.0	0.2	0.1	0.0	0.0	0.4	1.2	0.8	0.7	0.4	0.2	0.3	0.4		
	PrecStd	3.3	0.6	0.5	0.6	0.8	1.1	1.0	1.3	1.6	1.1	0.9	0.7	0.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.1	37.7	47.5	72.5	83.2	88.5	89.9	88.3	78.3	65.1	44.9	36.4		
		MCWB	32.9	33.4	39.0	56.3	62.5	70.6	72.0	71.5	63.9	54.1	41.2	33.8		
	2%	DB	30.2	34.2	41.6	64.3	78.7	84.2	85.4	82.0	72.1	59.1	40.7	33.4		
		MCWB	28.3	31.5	35.6	51.8	61.0	68.4	70.7	69.0	61.4	51.2	38.4	31.2		
	5%	DB	26.8	30.6	38.3	58.6	74.5	80.2	82.4	77.1	66.6	54.0	37.5	30.0		
		MCWB	25.3	28.1	33.5	48.1	59.3	67.1	69.6	66.7	58.6	48.4	35.4	28.3		
	10%	DB	22.9	27.0	35.7	54.0	69.7	76.0	79.1	73.0	61.8	50.0	34.8	26.7		
		MCWB	21.5	24.7	31.7	45.2	56.9	65.1	68.1	64.7	55.8	46.0	32.9	25.2		
	0.4%	WB	33.0	34.5	39.7	56.7	65.7	73.5	74.7	73.2	65.8	55.3	42.1	34.2		
		MCDB	34.9	37.0	45.9	69.1	77.1	84.3	85.3	85.4	74.2	62.0	44.4	35.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	28.7	31.7	36.4	52.5	62.6	70.3	72.6	70.1	62.1	51.6	38.9	31.4		
		MCDB	29.9	34.2	40.8	62.8	76.3	81.2	82.6	79.3	70.5	57.6	40.7	33.3		
	5%	WB	25.3	28.3	34.2	48.9	59.9	68.1	70.7	67.6	59.2	48.7	35.8	28.5		
		MCDB	26.8	30.4	37.7	57.6	72.4	78.6	80.3	75.5	65.8	53.7	37.4	29.9		
	10%	WB	21.5	24.7	32.1	45.7	57.3	65.9	68.7	65.3	56.4	46.1	33.1	25.2		
		MCDB	22.7	26.7	35.3	53.2	68.1	74.1	77.5	71.8	61.0	49.8	34.6	26.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	12.6	14.9	16.4	17.6	20.5	19.4	18.9	17.3	15.4	12.2	10.8	11.6		
		MCDBR	12.4	13.3	16.7	24.0	27.7	25.2	24.0	23.2	21.8	18.9	11.0	10.8		
		MCWBR	12.1	11.1	12.0	14.7	14.5	12.6	11.2	11.7	12.7	12.0	9.4	10.1		
		MCDBR	12.7	12.4	14.6	22.7	25.7	23.0	22.3	21.5	20.7	18.1	10.6	10.5		
	5% WB	MCWBR	12.3	10.5	11.1	14.5	14.5	12.8	11.5	11.6	12.6	11.9	9.4	10.0		
(40) Clear-Sky Solar Irradiance	taub		0.279	0.311	0.366	0.406	0.401	0.402	0.418	0.429	0.380	0.344	0.303	0.274		
	taud		2.036	2.034	1.953	2.093	2.260	2.318	2.273	2.214	2.314	2.277	2.134	1.995		
	Ebn at Noon		167	218	237	252	264	265	257	243	238	210	158	126		
	Edh at Noon		18	29	42	43	39	37	38	38	30	23	16	14		
(44) All-Sky Solar Radiation	RadAvg		194	495	953	1331	1615	1756	1708	1292	787	406	184	111		
	RadStd		17	39	57	128	141	125	145	126	78	62	26	15		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page